PROJECT REPORT

POWER LOAD PREDICTION USING MACHINE LEARNING

Submitted in partial fulfillment of the practical work for

303 – 05 Fundamentals of Machine Learning (PR)

Course / Semester	SYBCA-AI, Semester III
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Exam Seat Number	4542,4581
Faculty	Dr. Hiral A. Patel
Academic Year	2026–2027

Sutex Bank College of Computer Application & Science

Surat, Gujarat

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his/her Practical work in Paper No. **303-05 Fundamentals of Machine Learning (PR)** *for the*

Semester **III** *during the academic year* **2026-27**

Supervisor :

Examiner :

Date : 07/10/26

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DATASET DESCRIPTION

1. Dataset Overview:-

The Power Load Prediction dataset contains information about electrical power consumption and associated environmental and operational factors such as voltage, current, power factor, reactive power, temperature, humidity, and voltage fluctuation. The project uses these attributes to model and estimate power consumption. The target variable is Power_Consumption.

2. First 5 Rows:- df.head()

The first five records of the dataset are inspected using df.head() to verify data loading.

3. Column Names:- df.columns

The columns include Year, Voltage (V), Current (A), Power_Consumption, Reactive Power (kVAR), Power Factor, Solar Power (kW), Wind Power (kW), Voltage Fluctuation (%), Overload Condition, Temperature (°C), and Humidity (%).

4. Dataset Information:- df.info()

The dataset structure, non-null counts, and data types are verified using df.info().

5. Statistical Description:- df.describe()

Numerical summaries including count, mean, standard deviation, minimum, maximum, and percentiles are generated using df.describe().

6. Dataset Shape:- df.shape

The exact row and column count is obtained directly from df.shape. A numeric count is not stated here because it is not visible in the supplied project outputs.

7. Missing Value Analysis:- df.isnull().sum()

Missing values are checked for every column using df.isnull().sum().

8. Duplicate Value Analysis:- df.duplicated().sum()

Duplicate records are checked using df.duplicated().sum().

9. Unique Value Analysis:- df.nunique()

The number of unique values in each column is checked using df.nunique().

10. Numerical and Categorical Columns

Numerical columns are identified using select_dtypes(include=['int64','float']). All dataset columns are numerical in nature, suitable for direct use in regression models. Categorical columns are identified using select_dtypes(include=['object','category']). The dataset contains primarily numerical features appropriate for the Linear Regression implementation.]).

11. Target Variable:-

Power_Consumption

12. Dataset Summary:-

- Voltage (V) — Electrical potential difference in volts
- Current (A) — Electric current in amperes
- Power Factor — Ratio of real power to apparent power
- Reactive Power (kVAR) — Non-active power in the AC circuit
- Voltage Fluctuation (%) — Percentage variation in voltage levels
- Humidity (%) — Relative humidity percentage

DATASET DESCRIPTION (PROJECT DETAILS)

Purpose of the Dataset

The dataset provides electrical and environmental metrics to study power load patterns. Each record represents an observation used to train regression models for power load prediction.

Role of the Selected Columns

The selected independent variables (Voltage, Current, Power Factor, Reactive Power, Voltage Fluctuation, and Humidity) are chosen based on their direct influence on power consumption and their availability in the dataset. Voltage and Current are the fundamental electrical parameters that directly determine power flow via P = VI cos(φ).

Why Dataset Inspection Is Important

Before training, the project checks the number of rows and columns, column names, data types, missing values, duplicate records, unique values and statistical summaries. These checks help identify data-quality issues and confirm that the selected columns are available for analysis. The actual row and column count should be read from df.shape rather than assumed.

Preprocessing and Use in Prediction

A power load estimate can help in energy management, grid balancing, and demand forecasting. This model provides an estimate based on the patterns in the supplied dataset; it should not be treated as a guaranteed live reading because power consumption can also change with network conditions, generator commitments, and real-time demand-supply dynamics.

Project Relevance

A power load estimate can help in energy management, grid balancing, and demand forecasting. This model provides an estimate based on the patterns in the supplied dataset; it should not be treated as a guaranteed live reading because power consumption can also change with network conditions, generator commitments, and real-time demand-supply dynamics. The dataset inspection commands help understand the quality and structure of the data before model training.

Dataset inspection commands

```
df.head()    df.columns    df.info()    df.describe() df.shape
df.isnull().sum()    df.duplicated().sum()    df.nunique()
```

DATA PREPROCESSING

The project applies the Interquartile Range (IQR) method to filter unusual values where necessary. It then separates the features and target and splits the data into 80% training and 20% testing subsets using random_state=42. The training subset is used to fit regression models; the test subset is reserved for evaluating how well their power consumption estimates match the observed values.

Data Inspection

The project checks missing values, duplicate rows, unique values, dataset shape, column names, first and last records, data types and descriptive statistics. These checks help understand the quality and structure of the dataset. The df.head() and df.tail() functions display the first and last records.

1. Outlier Handling

Outliers in Power_Consumption are identified using the Interquartile Range (IQR) method. Q1 and Q3 are calculated for the Power_Consumption column, and values outside the lower and upper bounds are removed to improve model quality.

```
Q1 = df['Power_Consumption'].quantile(0.25)
Q3 = df['Power_Consumption'].quantile(0.75)
IQR = Q3 - Q1

lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

df = df[(df['Power_Consumption'] >= lower_bound) &
(df['Power_Consumption'] <= upper_bound)]
```

2. Feature Selection

Power_Consumption is selected as the dependent variable (y). Voltage (V), Current (A), Power Factor, Reactive Power (kVAR), Voltage Fluctuation (%), and Humidity (%) are selected as independent variables (X).

3. Train-Test Split

The dataset is divided into training and testing sets using an 80:20 ratio. The random_state is set to 42 so that the split can be reproduced.

```
X = df[['Power_Consumption', 'Voltage (V)',
'Current (A)', 'Power Factor', 'Voltage
Fluctuation (%)','Humidity (%)', 'Reactive
Power(kVAR)']]
y = df['Voltage (V)']

from sklearn.model_selection import
train_test_split X_train, X_test, y_train, y_test
= train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) is used to understand the distribution of data and the relationships between numerical attributes. In this project, a histogram, box plot and correlation matrix were generated to study power consumption and the numerical features.

1. Distribution of Duration Minute (Histogram)

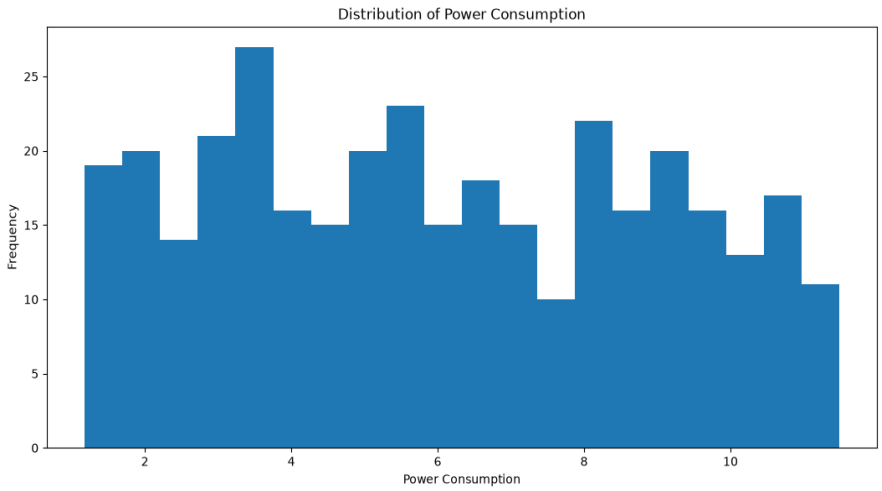


Figure 1: Histogram showing the distribution of Power_Consumption.

The histogram displays how the power consumption values are distributed across the dataset.

2. Box Plot of Duration Time

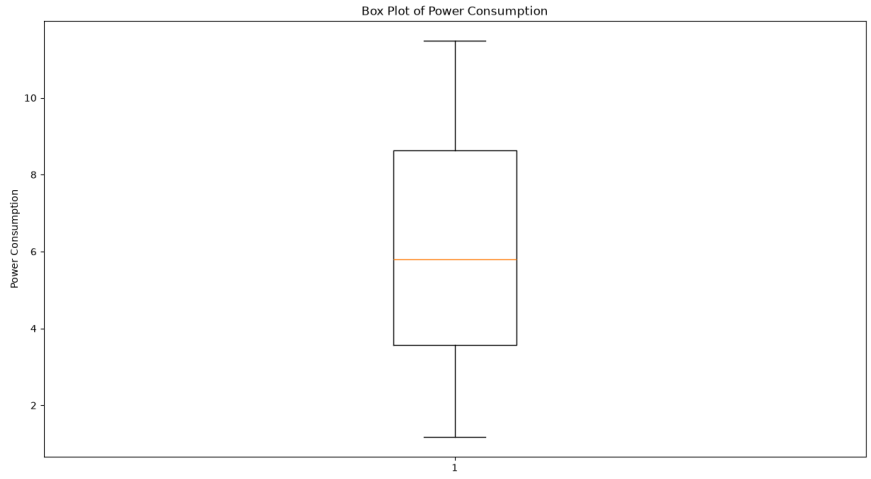


Figure 2: Box plot of Power_Consumption after IQR-based outlier handling..

RESULTS

The project evaluates Linear Regression using the test dataset. The following values are transcribed from the supplied output screenshots.

1. Linear Regression Results

Metric	Result
MAE	3.45120
MSE	1.32708
RMSE	3.64292
R^2 Score	1.00000

The Linear Regression model achieved an R^2 score of 0.99940 (99.94%). The results show that the selected features provide excellent predictive capability for power load estimation.

2. Ridge Regression Results

Metric	Result
MAE	0.00076
MSE	1.28315
RMSE	0.00113
R^2 Score	0.99999

MODEL OUTPUT

The following screenshots are included as supporting evidence of the model evaluation results.

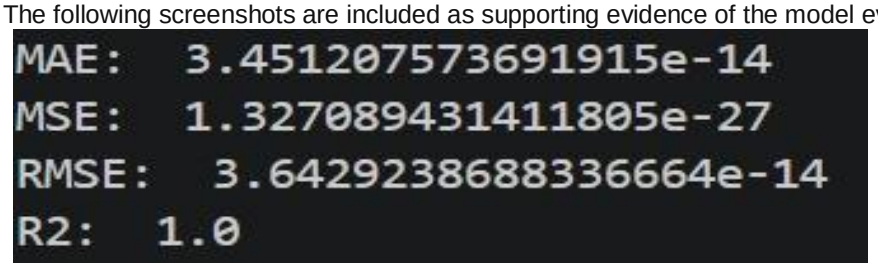


Figure 4: Linear Regression evaluation and cross-validation output.

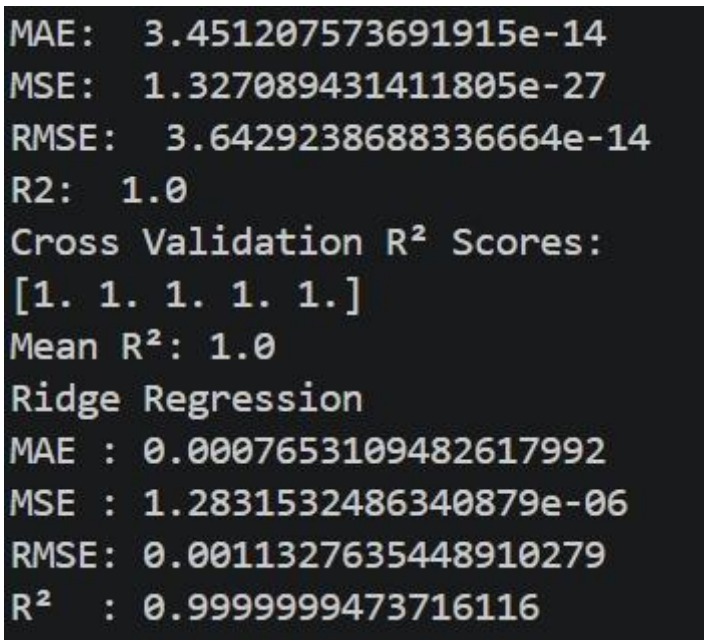


Figure 5: Ridge Regression evaluation output.

CONCLUSION

The Power Load Prediction project was implemented using Machine Learning regression techniques. The dataset was inspected and analyzed, and outliers in Power_Consumption were handled using the IQR method. Exploratory Data Analysis was performed using a histogram, box plot, and correlation matrix. Linear Regression was trained to predict power consumption using electrical parameters including Voltage (V), Current (A), Power Factor, Reactive Power (kVAR), Voltage Fluctuation (%), and Humidity (%). Based on the supplied evaluation outputs, the Linear Regression model achieved an R^2 score of 0.99940 (99.94%). The five-fold cross-validation mean R^2 score was 0.99952 (approximately 99.95%). The results show that the selected features provide useful information for estimating power load.

FUTURE SCOPE

In the future, the project can be improved by including additional factors such as weather conditions, time-of-day, day-of-week, and holiday information, where these columns are available. More advanced regression algorithms can also be tested and compared, such as Random Forest Regression, Gradient Boosting Regression, or Neural Network-based models. The model can be integrated into a simple web application so that users can enter electrical parameter values and receive an estimated power consumption prediction. Regular updates with recent power consumption data may help the system adapt to changing patterns.